

INTRODUCTION TO POWER BI DASHBOARDS

What is Power BI?

Power BI is a business analytics tool by Microsoft that allows users to:

- Import data from various sources
- Transform and model data
- Build interactive dashboards and reports
- Share insights across an organization

It is widely used for business intelligence, analytics, and data visualization.

Power BI Basics

1. Get Data

Import from Excel, SQL, CSV, Web, APIs, cloud services, etc.

2. Transform Data (Power Query)

Clean and prepare using:

- Remove duplicates
- Correct data types
- Replace values
- Split columns
- Merge queries
- Add calculated columns

3. Model Data

- Build relationships between tables
- Create measures using DAX (Data Analysis Expressions)
- Manage hierarchies (Year → Quarter → Month)

4. Build Visuals

Use charts, cards, slicers, maps, tables, etc., to visualize insights.

5. Publish & Share

Upload to Power BI Service, create dashboards, and share with others.

4. Understanding the Power BI Interface

Power BI Desktop has four main views:

View	Description
Report	Where you design dashboards and visuals
Data	View the dataset in table form
Model	Manage relationships between tables
Power Query Editor	Data cleaning and transformation

5. Common Power BI Visuals

Visual	Best for
Bar/Column Chart	Comparing values
Line Chart	Trends over time
Pie/Donut Chart	Proportions
Card KPI	Single summary value

Table/Matrix	Detailed data
Map	Location-based analysis
Slicer	Filters for user interaction

6. Basic DAX (Data Analysis Expressions)

DAX is Power BI's formula language.

Common examples:

- **SUM()** → total sales
- **COUNTROWS()** → count number of orders
- **IF()** → conditional logic
- **CALCULATE()** → apply custom filters

Example:

Total Sales = SUM(Orders[Sales])

7. Rules for a Good Dashboard

- ✓ Clear purpose
- ✓ Minimal clutter
- ✓ Consistent colors
- ✓ Uses interactive filters
- ✓ Align visuals properly
- ✓ Highlights KPIs and trends

8. Dashboard Design Best Practices

- Use 1–3 main colors

- Place KPIs at the top
- Put filters on the left or right side
- Avoid pie charts with too many slices
- Use tooltips instead of text-heavy visuals

LAB ACTIVITY

Objective:

Students will create an interactive dashboard showing Sales, Profit, and Category analysis using the Superstore dataset.

Difficulty: Beginner

Estimated Time: 1 hour

PART 1 — Import the Superstore Dataset

1. Download the Superstore dataset (Excel file).
 2. Open **Power BI Desktop**
 3. Click **Get Data** → **Excel**
 4. Select the Superstore file
 5. Load the following sheets:
 - Orders
-

PART 2 — Clean and Transform Data in Power Query

1. Go to **Transform Data**
2. In the *Orders* table:
 - Check data types (ensure Sales and Profit are decimal numbers)
 - Remove any blank rows
 - Split the *Order Date* column into:
 - **Year**
 - **Month Name**
3. Close & Apply

PART 3 — Create Basic Measures (DAX)

Open **Modeling** → **New Measure**, then add:

```
Total Sales = SUM(Orders[Sales])
Total Profit = SUM(Orders[Profit])
Total Quantity = SUM(Orders[Quantity])
Profit Margin = [Total Profit] / [Total Sales]
```

PART 4 — Build the Dashboard

A. KPI Cards

Add **3 Card visuals**:

- Total Sales
- Total Profit
- Profit Margin

Format:

- Color: Dark blue
- Title: White bold text

B. Sales by Category

Add a **Column Chart**:

- Axis: Category
- Values: Total Sales

Add a **Line Chart**:

- Axis: Order Date (by Month)
- Value: Total Sales
- Show trend line

C. Sales and Profit by Region

Add a **Filled Map**:

- Location: State
- Color: Total Sales

Add a **Bar Chart**:

- Axis: Region
- Value: Total Profit

D. Table for Detailed View

Add a **Table** visual with:

- Order ID

- Product Name
- Category
- Sales
- Profit
- Quantity

Enable conditional formatting:

- Profit: Turn negative values red

E. Add Slicers

Add slicers for:

- Category
- Region
- Year
- Month

Make sure they filter all visuals.

PART 5 — Final Dashboard Layout

Suggested Layout:

